Rates and Best Buys

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

KEY WORDS rate *(pāpātanga)* · unit rate *(pāpātanga kotahi)* · per · best buy · speed *(tere)* · distance *(tawhiti)*

A Unit rates ●○○

Remember: A rate compares two different quantities. "Per" means "for each one".

1. Find each rate.

a) \$12 for 4 kg: cost per kg = _____

b) 240 km in 3 hours: km per hour = _____

c) \$45 for 5 hours of work: pay per hour = _____

d) 600 words in 8 minutes: words per minute = _____

e) 18 L in 4 minutes: litres per minute = _____

f) \$7.20 for 6 bottles: cost per bottle = _____

g) 360 heartbeats in 5 minutes: beats per minute = _____

h) \$2.80 for 400 g: cost per 100 g = _____

B Using a rate ●○○

2. Answer each question.

a) A tap runs at 12 litres per minute. How much water runs out in 7 minutes?

b) Mere earns \$24.50 per hour. How much does she earn in 6 hours?

c) A car uses 8 litres of petrol for every 100 km. How much petrol does it use on a 350 km trip?

d) A printer prints 24 pages per minute. How long does it take to print 300 pages?

e) Mobile data costs \$15 per GB. Find the cost of 2.5 GB.

f) A heart beats 72 times per minute. How many times does it beat in 1 hour?

C Best buys ●●○

Remember: Find the price of one unit for each deal (the unit price), then compare.

3. Which is the better buy? Show the unit price for each.

a) Rice: 2 kg for \$5.60 or 5 kg for \$13.50

b) Juice: 1.5 L for \$3.90 or 2 L for \$5

c) Pens: 6 pens for \$4.50 or 10 pens for \$7.20

d) Mobile data: 4 GB for \$18 or 10 GB for \$42

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Speed, distance and time ●●○

> **Remember:** Speed = distance ÷ time. Distance = speed × time. Time = distance ÷ speed.

4. Find the speed.

a) 150 km in 2 hours **b)** 400 m in 50 seconds **c)** 100 m in 12.5 seconds

5. Find the distance travelled.

a) 60 km/h for 2.5 hours **b)** 12 m/s for 45 seconds **c)** 85 km/h for 3 hours

6. Find the time taken.

a) 240 km at 80 km/h **b)** 90 km at 60 km/h **c)** 600 m at 8 m/s

E Changing units of speed ●●●

> ⚠ **Watch out:** To change km/h to m/s, divide by 3.6. To change m/s to km/h, multiply by 3.6.

7. Change each speed to the other unit.

a) 72 km/h **b)** 15 m/s **c)** 90 km/h **d)** 10 m/s

F Problems ●●●

8. Mere drives 126 km in 1 hour 45 minutes. Find her average speed.

9. A cyclist rides at 24 km/h. How long does it take to ride 42 km? Give your answer in hours and minutes.

10. Petrol costs $2.85 per litre. A car uses 7.2 litres for every 100 km. Find the cost of petrol for a 250 km trip.

11. A 750 g box of cereal costs $6.30. A 1.2 kg box of the same cereal costs $9.60. Which is the better buy?

★ Extension

12. A train travels 180 km at 90 km/h and then 120 km at 60 km/h. Find its average speed for the whole journey.

13. Tap A fills a tank in 6 minutes. Tap B fills the same tank in 12 minutes. How long do they take to fill it together?

14. Sound travels at 340 m/s. Thunder is heard 4.5 seconds after the lightning flash. How far away is the storm, in kilometres?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Rates and Best Buys

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS rate *(pāpātanga)* · unit rate *(pāpātanga kotahi)* · per · best buy · speed *(tere)* · distance *(tawhiti)*

A Unit rates ●○○

Remember: A rate compares two different quantities. "Per" means "for each one".

1. Find each rate.

a) $15 for 5 kg: cost per kg = _____

b) 280 km in 4 hours: km per hour = _____

c) $54 for 6 hours of work: pay per hour = _____

d) 450 words in 6 minutes: words per minute = _____

e) 26 L in 4 minutes: litres per minute = _____

f) $9.60 for 8 bottles: cost per bottle = _____

g) 468 heartbeats in 6 minutes: beats per minute = _____

h) $4.50 for 600 g: cost per 100 g = _____

B Using a rate ●○○

2. Answer each question.

a) A tap runs at 15 litres per minute. How much water runs out in 6 minutes?

b) Hemi earns $26.50 per hour. How much does he earn in 8 hours?

c) A car uses 7 litres of petrol for every 100 km. How much petrol does it use on a 450 km trip?

d) A printer prints 32 pages per minute. How long does it take to print 400 pages?

e) Mobile data costs $12 per GB. Find the cost of 3.5 GB.

f) A heart beats 68 times per minute. How many times does it beat in 1 hour?

C Best buys ●●○

Remember: Find the price of one unit for each deal (the unit price), then compare.

3. Which is the better buy? Show the unit price for each.

a) Rice: 2 kg for $6.40 or 5 kg for $15.50

b) Juice: 1.5 L for $4.20 or 2 L for $5.40

c) Pens: 5 pens for $4.25 or 12 pens for $9.60

d) Mobile data: 5 GB for $22 or 12 GB for $51

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Speed, distance and time ●●○

> **Remember:** Speed = distance ÷ time. Distance = speed × time. Time = distance ÷ speed.

4. Find the speed.

a) 180 km in 3 hours **b)** 600 m in 75 seconds **c)** 200 m in 25 seconds

5. Find the distance travelled.

a) 70 km/h for 1.5 hours **b)** 15 m/s for 40 seconds **c)** 95 km/h for 4 hours

6. Find the time taken.

a) 270 km at 90 km/h **b)** 120 km at 80 km/h **c)** 900 m at 12 m/s

E Changing units of speed ●●●

> ⚠ **Watch out:** To change km/h to m/s, divide by 3.6. To change m/s to km/h, multiply by 3.6.

7. Change each speed to the other unit.

a) 54 km/h **b)** 20 m/s **c)** 108 km/h **d)** 25 m/s

F Problems ●●●

8. Tama drives 135 km in 2 hours 15 minutes. Find his average speed.

9. A cyclist rides at 20 km/h. How long does it take to ride 45 km? Give your answer in hours and minutes.

10. Petrol costs $2.90 per litre. A car uses 6.5 litres for every 100 km. Find the cost of petrol for a 400 km trip.

11. A 500 g box of cereal costs $4.40. A 1.2 kg box of the same cereal costs $10.20. Which is the better buy?

★ Extension

12. A train travels 240 km at 80 km/h and then 100 km at 50 km/h. Find its average speed for the whole journey.

13. Tap A fills a tank in 10 minutes. Tap B fills the same tank in 15 minutes. How long do they take to fill it together?

14. Sound travels at 340 m/s. Thunder is heard 6.5 seconds after the lightning flash. How far away is the storm, in kilometres?

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.